Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 5 Tutorial Worksheet: Routing Protocols and Routing Attacks

Field	Detail
Based on	Week 5, Routing Protocols and Routing Attacks lecture (protocol fundamentals, five routing attacks, and defenses)
Learning Outcomes	LO1 (describe network vulnerabilities/attacks) and LO2 (evaluate mitigation techniques)
Format	12 structured questions, no MCQs. Answer in your own words.
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic. Use them directly when revising for the mid-semester test and final exam.

These questions are drawn directly from this week's lecture. Several ask you to compare or evaluate rather than just describe. That's deliberate: LO2 is specifically about weighing mitigation options, not just naming them. Write full answers, not one-line definitions.

Routing Protocol Fundamentals

Q1. Compare RIP, OSPF, EIGRP, and BGP by type (distance-vector, link-state, advanced distance-vector, or path-vector) and by the metric each one uses to select a best path. *[LO1]*

Q2. Describe the message types OSPF uses to form and maintain a neighbor relationship, and explain which specific message type MD5 authentication actually has to protect. *[LO1]*

Q3. Explain why BGP's transport (TCP, port 179) is fundamentally different from the other three protocols covered this week, and what that difference means for how BGP's own authentication actually works. *[LO1]*

Routing Attacks

Q4. Explain how Routing Table Poisoning works, and describe how the exact technique differs between a distance-vector protocol like RIP and a link-state protocol like OSPF. *[LO1]*

Q5. Distinguish Unauthorised Route Injection from Routing Table Poisoning. What does the attacker actually gain in each case that's different from the other? *[LO1]*

Q6. Explain why Route/Prefix Hijacking is considered a BGP-specific attack, using the 2008 YouTube hijack to illustrate your answer. *[LO1, LO3]*

Q7. Explain the difference between a blackhole outcome and a sinkhole outcome, and why both share the same underlying vulnerability as the injection and poisoning attacks. *[LO1]*

Defenses

Q8. A network administrator asks you which single defense would do the most to reduce risk across this week's five attacks. Evaluate neighbor authentication (MD5) against this question, including what it does not protect against. *[LO2]*

Q9. Explain why this week's lab only covers OSPF and EIGRP authentication, not all four protocols covered in the lecture. *[LO2]*

Q10. Compare how MD5 authentication is actually configured on OSPF versus EIGRP, and explain the practical trade-off between the two approaches. *[LO2]*

Q11. A colleague says: 'We've applied a distribute-list on our OSPF network, so we're now confident no unauthorised route can propagate to other routers.' Evaluate this claim. Is it accurate? Why or why not? *[LO2]*

Q12. Explain what RPKI does, and why it directly addresses the specific weakness that caused the 2008 YouTube hijack. *[LO1, LO2]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. This week's lab is direct hands-on practice for OSPF and EIGRP authentication. Configuring it yourself will cement the theory faster than rereading it.
3. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1 and LO2) and final exam (MCQ + structured essay, LO1 through LO5). The evaluation-style questions above are close in format to the structured-essay section.